Tina Zhao

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EDUCATION

Teachers College, Columbia University **September 2024 - February 2026**
M.S. in Learning Analytics | GPA: 3.92/4.00 | New York, NY

University of Manchester **September 2021 - June 2024**
B.Sc. in Education | Upper Second-Class Honours (2:1) | Manchester, United Kingdom

RESEARCH EXPERIENCE

Research Associate | University of California, Irvine Digital Learning Lab **March 2026 - Present**
Principal Investigator: Mark Warschauer | Irvine, CA

Computing & AI for All - NSF-Funded Research-Practice Partnership

- Conduct research on how K-12 curriculum and teacher professional learning can support AI literacy, computational thinking, and equitable participation; synthesize teacher workshops, interviews, and implementation meetings into iterative revisions of lessons, scaffolds, standards alignment, platform guidance, and the public curriculum portal.
- Designed a prompt-level study of 2,433 student messages from LLM-supported block-based programming; defined the unit of analysis, developed the codebook, completed a 600-message reliability pilot (84.5% agreement; Cohen's kappa = .814), and refined decision rules through disagreement analysis.

Science Inquiry Studio

- Co-initiated an inquiry-first design-research strand integrating AI into K-12 science learning while maintaining students' active role in questioning, reasoning, and investigation; translate early-stage concepts into learning-design principles, interactive science investigations, teacher guides, and digital prototypes.

Agentic Coding Studio - NSF-Funded Research & Design Project

- Develop curriculum and instructional materials for learner-directed agentic coding, in which students collaborate with an AI coding agent while retaining responsibility for goal setting, plan review, delegation decisions, testing, debugging, revision, and explanation of design choices.
- Design and refine early-stage CreatiCode XO learning experiences, including the Build Breakout sequence, and develop student-facing scaffolds and teacher supports that move learners from tool fluency toward increasingly independent planning, evaluation, revision, and creative control.

Teacher Professional Learning & Research-Practice Implementation

- Designed and facilitated multi-day Computing+AI professional development across an in-person Hawaii cohort (15 teachers) and a five-day online cohort (~10 teachers); developed instructional materials, supported AI literacy and coding activities, iteratively revised resources in response to teacher needs, and represented the project at CSTA 2026 through curriculum demonstrations and recruitment.

Research Associate | Learning Analytics Lab, Teachers College, Columbia University **December 2025 - June 2026**
Human-AI Co-Design for Research Question Formulation | Principal Investigator: Yasemin Gulbahar

- Designed and conducted an IRB-approved, within-subjects think-aloud study with six master's students completing four sessions each across two learning analytics scenarios and two LLMs (24 sessions).
- Built a dual analytic framework: a Product rubric assessed specificity, data-method alignment, theoretical depth, and inquiry coherence, while a Process rubric captured prompt iteration, critical evaluation, help-seeking, and challenge resolution.
- Segmented chat transcripts and think-aloud data into 129 goal-centered interaction episodes; applied hybrid deductive-inductive coding with documented decision rules and manually adjudicated final coding and rubric scores across 24 Research Inquiry Clusters.
- Integrated product quality with interaction trajectories to examine design moves such as constraint checking, construct operationalization, method verification, and correction of infeasible AI suggestions; translated the analysis into a practice-based roadmap and submitted a first-authored AERA 2027 individual paper proposal.

Graduate Research Project | LLM-Assisted Annotation of Online Learning Comments **January 2025 - May 2025**
Teachers College, Columbia University | Course Instructor: Renzhe Yu

- Evaluated an OpenAI API-based multi-label annotation workflow using 289 comments from 124 MIT OpenCourseWare videos across 10 course playlists and 10 comment categories; benchmarked zero- and few-shot annotations against two human coders using category-level Cohen's kappa and analyzed disagreement across prompting conditions.

Undergraduate Researcher | Teaching Excellence Framework and Lecturer Practice **October 2023 - June 2024**
Undergraduate Thesis | University of Manchester

- Designed and conducted a qualitative dissertation study with five university lecturers across Education, Climate Change, Circular Economy, and Fashion Management; developed a semi-structured interview protocol and coded transcripts into four thematic areas examining how TEF influenced teaching motivation, pedagogical decisions, and perceptions of teaching quality.
- Found limited direct influence of TEF on classroom practice, alongside mixed views on feedback-driven improvement and concerns about metric-based accountability, bias in student evaluations, and the marketization of higher education.

RESEARCH OUTPUTS AND WORK IN PROGRESS

Zhao, T., & Gulbahar, Y. (2027). Human-AI co-thinking in research question formulation as methodological apprenticeship. Individual paper proposal submitted to the AERA 2027 Annual Meeting. Under review.

Zhao, T., Rodriguez, E., Saito Stehberger, D. E., Yu, B., Esalomi, E., & Yeh, A. (2027). From classroom to curriculum: How teacher co-design shaped an AI-enhanced coding curriculum. Manuscript in preparation for submission to the 17th Symposium on Educational Advances in Artificial Intelligence (EAAI-27). In preparation.

Zhao, T., Rodriguez, E., Saito Stehberger, D. E., Yu, B., Esalomi, E., & Yeh, A. (2027). From explanation to agentic coding: Build Breakout with CreatiCode XO. Session proposal in preparation for CALIE Spring Forward 2027. In preparation.

LEARNING DESIGN AND TEACHING EXPERIENCE

Think Academy | Mathematics Curriculum Designer & Instructor **March 2026 - Present**

- Design and teach individualized mathematics lessons for students in Grades 2-6 aligned with Common Core State Standards (CCSS), translating grade-level concepts into developmentally appropriate explanations, visual representations, and scaffolded problem-solving activities.
- Adapt instructional sequences and scaffolds based on students' reasoning, misconceptions, and formative performance, emphasizing conceptual understanding, explanation of mathematical relationships, and transfer to new contexts.

ReachOut Society | After-School Activity Designer and Educational Mentor **2023 - 2024**

Undergraduate Placement | Manchester, United Kingdom

- Designed and facilitated after-school learning activities in low-income, culturally and linguistically diverse school communities; used formative observation and performance data to adapt small-group curricula and targeted supports for lower-performing and multilingual learners.
- Collaborated with school staff to refine instructional scaffolds, monitor student progress, and provide targeted academic and social support.

NORY | STEM Instructor and Curriculum Designer **May 2025 - August 2025**

New York, NY

- Designed and taught inquiry-based K-5 STEM lessons using robotics, coding, and collaborative problem solving; adapted activities and scaffolds to students' developmental needs and formative observations.

RESEARCH METHODS AND TOOLS

Interaction and qualitative analysis: Think-aloud protocols; segmentation and coding of student-LLM interactions; hybrid deductive-inductive codebook development; inter-rater reliability and disagreement adjudication (Cohen's kappa).

Design and implementation research: Design-based research; curriculum and learning-environment design; teacher professional learning; research-practice partnership implementation; classroom observation and iterative refinement.

Quantitative and computational analysis: Python, R, SQL, and SPSS for regression, process-data analysis, and educational data mining; OpenAI API for structured text annotation.